Evaluating Potential Bias in MLB Prospect Rankings

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Introduction:

Evaluating young talent is a pivotal, but also incredibly difficult, part of building a successful Major League Baseball (MLB) organization. Prospects vary significantly in age, position, developmental background, physical characteristics, and access to resources, making it difficult to accurately predict their future performance. MLB organizations dedicate substantial resources to identifying and developing young players, yet even highly ranked prospects frequently fail to meet expectations. This raises questions about how accurately prospect rankings predict future performance and whether certain types of players are systematically overvalued or undervalued.

Sims and Addona (2016) investigated how age influences MLB draft selections and subsequent performance, finding evidence that younger high school players were systematically undervalued which raises further questions about whether similar patterns exist in prospect rankings after players enter professional baseball. Gentile (2022) examined MLB's academy system in the Dominican Republic, highlighting structural inequalities in the treatment and development of international prospects. These findings suggest that players entering professional baseball through different developmental pathways may face different opportunities and expectations, potentially influencing how they are evaluated. Furthermore, modern prospect evaluations rely on a combination of subjective scouting grades and objective statistics,

including physical characteristics and measurable performance metrics. FanGraphs' The Board evaluates prospects using Future Value (FV) ratings, scouting grades, and other information intended to estimate their future contributions. However, these evaluations may place greater emphasis on certain characteristics than others. For example, players with impressive physical tools may receive more favorable evaluations than players who rely on functional baseball skills such as command, bat control, or pitch selection. Our project investigates whether FanGraphs' prospect evaluations systematically misjudge players based on developmental background, physical characteristics, and scouting attributes. Specifically, we examine whether certain groups of prospects consistently outperform or underperform expectations based on their initial FV ratings. By comparing prospect evaluations with subsequent MLB performance, we hope to identify potential patterns of bias and better understand the limitations of predicting future player success.

As baseball players who closely follow the sport, our interest in this topic stems directly from watching this tension play out on the field. We frequently see highly touted prospects struggle while under-the-radar players thrive, making us question if modern scouting's obsession with raw, measurable data causes evaluators to overlook fundamental baseball qualities. By comparing FanGraphs' initial FV ratings with subsequent major league performance, our project investigates whether the industry systematically misjudges players based on their developmental backgrounds or physical profiles.

Ultimately, our goal is to identify potential patterns of bias to better understand the limitations of modern scouting, particularly regarding how flawed evaluations might restrict the opportunities and resources available to young athletes.

Research Questions:

1. Systemic Background Biases: Do FanGraphs' evaluations systematically misvalue players based on demographic factors such as age and developmental background (high school, college, or international signing)? How do signing bonuses and Risk classifications relate to these differences, and are they associated with evaluation accuracy?
2. Biometric Archetypes: Do FanGraphs' evaluations exhibit systematic bias toward certain physical archetypes, such as height and weight? Do players who fall outside traditional physical expectations consistently outperform their projections?
3. "Stuff" vs. "Feel" Valuation: Do FanGraphs' evaluations overvalue raw, measurable tools, such as fastball velocity and exit velocity, at the expense of functional baseball skills, such as command, bat control, and pitch selection?

Expectations:

We expect that higher-rated prospects will generally achieve greater MLB success, although we anticipate considerable variation in individual outcomes. Based on previous research, we hypothesize that age and developmental background may influence evaluation accuracy, with younger prospects and international signees potentially being undervalued. We also expect that international players will receive higher Risk classifications than domestic players and that signing bonuses may influence evaluations beyond what is justified by future performance.

For our second research question, we hypothesize that players who fall outside traditional physical archetypes will be more likely to outperform their initial projections.

Finally, we expect that FanGraphs' evaluations will place greater emphasis on measurable physical tools, while certain functional baseball skills may be more closely associated with subsequent MLB performance.

Data and Data Preparation:

To address our research questions, we analyzed FanGraphs' The Board prospect rankings from 2020 through 2023, combined with MLB Wins Above Replacement (WAR) data from 2020 through 2026. Our original dataset contained 4,896 observations and 53 variables, including player characteristics, prospect evaluations, scouting grades, and MLB performance. Because some players appeared in multiple ranking years, we cleaned the dataset to retain one observation per player, using their most recent available ranking. This reduced our dataset to 2,450 prospects. Our dataset includes general player characteristics such as age, position, height, weight, developmental background, and signing bonus. We also included FanGraphs' Future Value (FV) ratings, Risk classifications, and individual scouting grades. These variables allow us to investigate whether certain characteristics are associated with differences between projected and actual MLB performance. We performed several cleaning steps to prepare the data for analysis. First, we standardized variables that were not originally stored in numerical formats. We converted height measurements from feet and inches to inches and signing bonuses into numerical dollar amounts. We also converted FV ratings into numerical values and separated scouting grades containing current and projected evaluations, retaining the projected grade for each tool. Next, we created several variables to address our research questions. We classified players into three primary developmental backgrounds: High School, College, and International, using information about where and how they originally signed. We also grouped positions into

broader categories, including pitchers, catchers, corner infielders, middle infielders, and outfielders. To examine physical archetypes, we created Height Build and Weight Build variables based on position-specific averages and standard deviations. Players more than one standard deviation above or below the average for their position group were classified as atypical, while players within that range were classified as typical. This allowed us to compare physical characteristics relative to positional expectations rather than using the same standards for every player. Finally, we created variables measuring MLB performance. We calculated Total WAR by summing each player's seasonal WAR values and identified Seasons Played as the number of seasons in which a player recorded nonzero WAR. Average WAR was then calculated by dividing Total WAR by Seasons Played, with players who had no nonzero-WAR seasons assigned an Average WAR of zero.

Our final cleaned dataset contained 2,450 prospects and 45 variables. Although the full dataset was used for some analyses, the number of observations varied across models depending on the availability of scouting information, signing bonuses, and other explanatory variables.

Methods:

Our three research questions investigate whether FanGraphs' prospect evaluations make systematic errors for certain types of players. To answer these questions, we used ordinary least squares (OLS) regression.

Rather than focusing exclusively on predicting MLB performance, we wanted to determine whether particular player characteristics were associated with outperforming or underperforming an initial FV evaluation. We first fit a simple linear regression predicting Average WAR from FV. We then calculated the residual for each player by subtracting the

model's predicted Average WAR from the player's observed Average WAR. A positive residual indicates that a player performed better than predicted by the baseline model, while a negative residual indicates that he performed worse. This allowed us to use the difference between observed and predicted performance as the outcome in subsequent models. If players with certain characteristics consistently had positive residuals, that would suggest they were outperforming the expectations represented by their FV grades. We selected OLS regression because WAR residuals are continuous and the resulting coefficients are relatively straightforward to interpret. Each coefficient represents the estimated difference in WAR residuals associated with a particular characteristic, holding the other included variables constant. We also included age in all of our models because it is related to developmental background, signing bonuses, and the amount of time players have had to reach the major leagues.

Insert visual/table of FV predicting WAR

For our first research question, we examined the relationships between developmental background, age, signing bonuses, Risk classifications, and WAR residuals. We began by modeling signing bonuses as a function of developmental background, age, and position. Because signing bonuses were heavily right-skewed, we transformed them using the natural logarithm of bonus plus one. Next, we constructed regression models predicting WAR residuals from developmental background and age. We then added signing bonuses to investigate whether financial investment was associated with evaluation accuracy. After examining these relationships, we introduced an interaction between developmental background and signing bonus. This allowed us to test whether the relationship between signing bonuses and WAR residuals differed for high school, college, and international prospects. Finally, we examined

whether background, age, and signing bonuses were associated with Risk classifications and whether those classifications predicted FV and WAR residuals.

For our second research question, we investigated whether height and weight were associated with systematic differences in evaluation accuracy. We constructed four regression models, controlling for age and position group throughout because physical builds and developmental trajectories vary across positions. First, we predicted WAR residuals using our categorical Height Build and Weight Build classifications, and fit a parallel model predicting initial Future Value (FV) to see how physical archetypes and positions shaped baseline grades. Next, we used continuous height and weight measurements to test whether our residual findings depended on categorical definitions. Finally, we added an interaction between height and weight categories to examine whether specific combinations of physical traits were associated with evaluation errors.

For our third research question, we separated hitters and pitchers because the two groups are evaluated using different scouting attributes. For hitters, we predicted WAR residuals and then fit a parallel model predicting initial Future Value (FV). We used scouting grades for Raw Power, Game Power, Hit, Speed, Fielding, Pitch Selection, and Bat Control. We then constructed a second model pairing objective measurables (maximum exit velocity and hard-hit rate) with functional skill grades (Hit, Bat Control, and Pitch Selection), controlling for position group and age throughout. For pitchers, we predicted WAR residuals using Fastball, Command, Changeup, Slider, and Curveball grades, followed by a second model combining top fastball velocity and breaking-ball spin rate with Command grades, controlling for age in both. Together, these models tested whether raw tools or functional game skills were associated with systematically exceeding or falling short of FV-based expectations.

Results:

Future Value (FV) rating was a statistically significant predictor of average WAR ($\beta=0.0571$, $p < 0.001$). This means 10 points of FV correlates to roughly 0.5 WAR per season. These predictions of WAR by FV were then used to create residuals which followed a right-skewed distribution with an extreme peak around 0 and a truncated left tail.

From our models for our first question a player's age and background are statistically significant predictors of their bonus (controlling for position group). Age has a negative correlation with bonus, and compared to a standard player coming out of college, highschool players make more ($\beta=0.3708$, $p<0.001$) while international players make much less ($\beta=-1.2969$, $p<0.001$). The position group correlated with the lowest bonus was pitchers. The only consistently statistically significant feature across all of our models predicting residuals was age ($\beta=0.0148$ to 0.0370, $p<0.001$ to 0.048). This feature always indicated overperformance. However when we added the interaction term Background * log_bonus the coefficients for international background and international background*log_bonus were both statistically significant ($\beta=0.9124$, $p<0.001$ and $\beta=-0.0661$, $p<0.001$). Age ($\beta=-0.1205$, $p<0.001$) and being from a high school background ($\beta=0.2015$, $p<0.001$) or international background ($\beta=0.2285$, $p<0.001$) were all statistically significant predictors of risk classification. Signing Bonus was not a significant predictor ($\beta=-0.0004$, $p=0.967$). Risk was not associated with WAR residuals (Low Risk: $\beta=0.0096$, $p=0.828$; Med Risk: $\beta=0.0237$, $p=0.483$) but was a significant predictor of FV (Low Risk: $\beta=1.4137$, $p<0.001$; Med Risk: $\beta=1.2664$, $p<0.001$). Age was notably not a significant predictor of FV ($\beta=0.0587$, $p=0.417$).

As far as physical archetypes the only height or weight build that showed statistical significance with WAR residuals was the typical weight build ($\beta=-0.0700$, $p=0.033$). However,

when holding height and weight constant the corner infield position group did seem to consistently underperform their projections (β=-0.1842, p=0.002). While most height and weight builds did not predict WAR residuals they did predict FV (Tall: β=1.0296, p=0.013, Typical Height: β=0.2386, p=0.440, Light: β=-1.4910, p<0.001, Typical Weight: β=-0.6418, p=0.039). Notably, the position group corner infield did not predict FV (β=-0.3757, p=0.508). None of the interaction terms between height build and weight build reached statistical significance.

For how grades predict WAR residuals, for hitters the speed tool and hit tool were predicted overperformance (β=0.0051, p=0.043 and β=0.0083, p=0.048). These were not the tools that were the strongest correlated with FV though as that was game power (β=0.4370, p<0.001) which notably was not a significant predictor of WAR residuals (β=0.0081, p=0.098). Positionally, catchers (the baseline group) were correlated with the highest FV as every other position had a significant negative coefficient. For pitchers, fastballs and then breaking balls were significant predictors of WAR residuals. Fastball grades were correlated with overperformance (β=0.0092, p=0.009) while breaking balls were correlated with underperformance (Curveball: β=-0.0099, p=0.030, Slider: β=-0.0092, p=0.042). None of the measurables for pitchers or hitters were statistically significant.

Discussion:

Our findings reveal notable systematic biases in how prospects are evaluated. While FV is a reliable predictor of baseline WAR, with every 10 points translating to roughly 0.5 WAR per season, the residuals from these projections highlight specific areas where evaluators consistently over or under-value player traits.

One of the most clear trends throughout all of our models was the relationship between age and WAR residuals. Because demographic backgrounds are inherently tied to specific age brackets (international prospects typically sign at 16-18, high schoolers at 17-19, and college players at 20-22) evaluators effectively use background as a proxy for youth in their baseline projections. In the initial FV model, numerical age itself is not statistically significant. Instead, the algorithm artificially inflates the Future Value of high school and international prospects based purely on the perceived ceiling of their younger demographic.

However, when analyzing actual Major League performance against these algorithmic expectations (WAR residuals), the predictive power of background evaporates, and numerical age emerges as the only consistently significant positive predictor of overperformance. This reveals a stark algorithmic ageism. The young, highly-touted international and high school prospects struggle to meet the inflated expectations set by their demographic proxies. Conversely, because older players are systematically under-projected and penalized in FV calculations, they consistently generate positive WAR residuals. By discounting the realized, refined skills of older prospects in favor of demographic ceilings, the algorithm practically guarantees that older players will outperform their artificially lowered ratings. Interestingly this seems to contradict the findings of Sims and Addona (2016), who found that younger prospects were undervalued.

This demographic bias fundamentally distorts the sport's economic structure. Younger prospects routinely command significantly higher signing bonuses under the industry-wide assumption that they possess higher developmental ceilings. While this makes sense in principle our residual analysis proves this ceiling disparity is not reflected in actual Major League production. Because older players systematically overperform their baseline projections, the

current distribution of signing bonuses is highly unfair, disproportionately rewarding unfulfilled demographic ceilings rather than compensating for the surplus WAR consistently generated by older prospects.

This signing bonus inequality specifically impacts international prospects. As seen by the statistically significant interaction term with a negative coefficient, compared to college and high school prospects, international prospects with lower signing bonuses tend to overperform their FV projections while prospects with higher singing bonuses tend to underperform. One explanation for this is the labor exploitation that takes place in these international markets as detailed in research like Gentile (2022). Well known more highly-touted prospects are forced to be compensated fairly but lesser known but nonetheless very talented prospects with very little bargaining power are given low bonuses and grades which they then overperform.

Insert signing bonus by age chart

Physical archetypes play an overimportant role in generating prospect ratings, pointing to a biometric bias where rewarding the "ideal baseball body." Models confidently penalize light builds and reward tall builds when calculating FV, yet these physical archetypes generally fail to predict WAR residuals. This suggests that while "look bias" is heavily programmed into the initial valuations, it does not actually drive performance deviations, except for the "typical weight" build, which correlates with underperformance.

There also appears to be a flaw in how positions are graded. The corner infield position group consistently underperformed their WAR projections. This is likely because scouts over-value corner infielders offensive potential without fully taking into account the harsh defensive penalty (a quality of WAR) their position possesses causing them to quickly lose value

if their bat doesn't pan out. On the other end of the spectrum compared to other positions and holding tool grades constant, catchers overperform their predictions. This is likely due to the slow developmental curve of catching and its very high defensive positional adjustment. These two cases seem to suggest that the positional component of WAR is not something that evaluators entirely take into account which is very debatable. There are many proponents of this positional adjustment score, while there are also people the don't think it reflects true in game value. Overall it is difficult to fully say that these positional misevaluations actually reflect bias.

While objective, measurable metrics were significant predictors of FV, they were not correlated with WAR residuals. Things like maximum fastball velocity and exit velocity are flashy metrics that contribute a lot to the hype around a prospect, and also certainly do have some bearing on that prospect's success. Our models showed that these characteristics were properly valued by FV. The tools that were undervalued or overvalued were more subjective, less flashy things. For hitters the tool that correlated the most to FV was Game Power which was insignificant at predicting WAR residuals and is therefore properly rated by the model. However the hit tool and speed tools were significant when predicting WAR residuals and indicated overperforming projections. These are less flashy tools and the players that possess them tend to fly under the radar and are systematically undervalued. For pitchers the same can be said. Maximum velocity was not a significant predictor of WAR residuals suggesting that it is properly rated. However, fastball grades themselves indicated overperformance, meaning that the less obvious characteristics that make a good fastball such as deception and movement profile were also systematically undervalued.

One of the more surprising things from our analysis was the fact that Fangraphs risk projections seemed to be relatively accurate. While risk classifications heavily dictate the initial

FV, with low and medium risk driving higher baseline projections, risk has absolutely no statistical association with WAR residuals. This indicates that the models are successfully pricing risk into the baseline without bias and the uncertainty of a prospect's profile is accurately neutralized in the FV grade, preventing it from skewing the residual outcomes. Risk is unequally distributed across demographic groups but that distribution does seem to correlate to actual production.

Limitations and Uncertainty:

Several limitations affect how confidently we can interpret our results.

First, our analysis uses Average WAR as the primary measure of player performance. Although WAR provides a useful standardized metric, it does not capture every aspect of a player's development or contribution. Playing time also affects opportunities to accumulate WAR, and playing time may itself be influenced by a player's prior reputation or prospect ranking.

Second, our dataset includes prospects evaluated between 2020 and 2023. Many of these players are still developing, meaning their observed MLB performance may not represent their eventual career outcomes. This limitation is especially important for younger prospects, who may require additional years before reaching the major leagues.

Third, our models used different sample sizes because scouting information was unavailable for certain players. Models involving detailed scouting grades contained considerably fewer observations than our models examining general player characteristics.

Consequently, some models may disproportionately represent prospects who received more detailed scouting evaluations.

Fourth, our WAR residuals were strongly right-skewed, with many players producing little MLB value and a relatively small number substantially exceeding expectations. This distribution raises concerns about the reliability of conventional OLS standard errors and statistical significance tests. Before the final report, we plan to investigate whether our conclusions remain consistent when using robust standard errors. Our baseline model also assumes a linear relationship between FV and Average WAR. If the relationship is nonlinear, our residuals may contain patterns that are not fully explained by the player characteristics we examined.

Fifth, categorizing developmental backgrounds strictly into high school, college, and international groups oversimplifies the diverse pathways players take to professional baseball. This broad classification groups together players with vastly different developmental experiences—such as four-year university athletes versus junior college players, or international signees with differing levels of access to elite academy resources. By condensing these pathways into three rigid categories, our models may obscure more nuanced trends in how specific, localized developmental contexts influence evaluation accuracy.

Finally, because our research relies on observational data, we cannot establish that any particular characteristic causes a player to be overvalued or undervalued. For example, although older players tended to outperform their evaluations, this does not demonstrate that age itself causes better performance. Other factors related to age, development, or opportunities may contribute to the observed relationship.

Conclusion:

By analyzing FanGraphs' Future Value (FV) projections against actual Major League production, this research shows that while baseline algorithms effectively price risk and "loud" tools like power and velocity, they systematically misprice less glamorous attributes. The evaluations have elements of algorithmic ageism, artificially inflating the grades of younger high school and international prospects based purely on demographic expectations. This creates massive economic inefficiencies in signing bonuses and leaves older prospects systematically undervalued. There is also evidence of a biometric "look bias" that heavily influences early grading based on an ideal physical frame, even though those traits do not actually predict future performance deviations.

Ultimately, by addressing these inequalities and inefficiencies, prospect evaluators can identify substantial surplus value that the rest of the industry ignores.

What still need to be done:

- Polish up writing, expand more on discussion, conclusion and limitations and condense earlier sections
- Connect analysis more to questions
- Add figures
- Try to address some of these limitations although that would probably involve going back to source data which could be a large undertaking
- Clean up Github